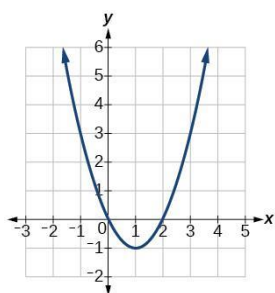
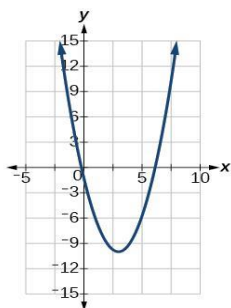


Section 3.2 Exercise Answers

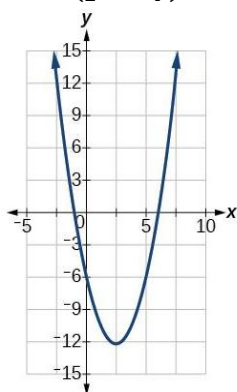
1. $f(x) = (x - 6)^2 - 4$; $(6, -4)$
2. $f(x) = (x + 1)^2 - 4$; $(-1, -4)$
3. $f(x) = \left(x - \frac{1}{2}\right)^2 - \frac{1}{4}$; $\left(\frac{1}{2}, \frac{1}{4}\right)$
4. $f(x) = \left(x + \frac{5}{2}\right)^2 - \frac{33}{4}$; $\left(-\frac{5}{2}, -\frac{33}{4}\right)$
5. $f(x) = 2(x + 2)^2 - 18$; $(-2, -18)$
6. $f(x) = 3(x - 1)^2 - 12$; $(1, -12)$
7. $f(x) = 2\left(x - \frac{3}{2}\right)^2 - \frac{9}{2}$; $\left(\frac{3}{2}, -\frac{9}{2}\right)$
8. $f(x) = 3\left(x - \frac{5}{6}\right)^2 - \frac{37}{12}$; $\left(\frac{5}{6}, -\frac{37}{12}\right)$
9. Minimum: $\left(-\frac{5}{2}, -\frac{1}{2}\right)$; $x = -\frac{5}{2}$
10. Minimum: $\left(\frac{5}{2}, -\frac{17}{2}\right)$; $x = \frac{5}{2}$
11. Maximum: $(2, 7)$; $x = 2$
12. Minimum: $\left(-\frac{1}{8}, -\frac{17}{16}\right)$; $x = -\frac{1}{8}$
13. Maximum: $\left(\frac{3}{4}, \frac{5}{4}\right)$; $x = \frac{3}{4}$
14. Minimum: $\left(-3, -\frac{7}{2}\right)$; $x = -3$
15. Maximum: $(-3, 6)$; $x = -3$
16. Domain: $(-\infty, \infty)$; Range: $[2, \infty)$
17. Domain: $(-\infty, \infty)$; Range: $(-\infty, -6]$
18. Domain: $(-\infty, \infty)$; Range: $[-5, \infty)$
19. Domain: $(-\infty, \infty)$; Range: $[0, \infty)$
20. Domain: $(-\infty, \infty)$; Range: $[-12, \infty)$
21. Vertex: $(1, -1)$; y -intercept: $(0, 0)$; x -intercepts: $(0, 0), (2, 0)$



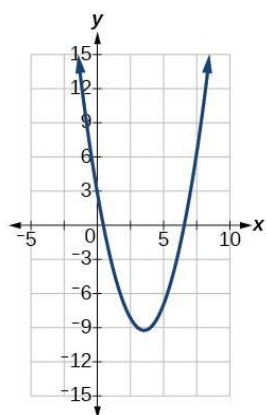
22. Vertex: $(3, -10)$; y -intercept: $(0, -1)$; x -intercepts: $(3 \pm \sqrt{10}, 0)$



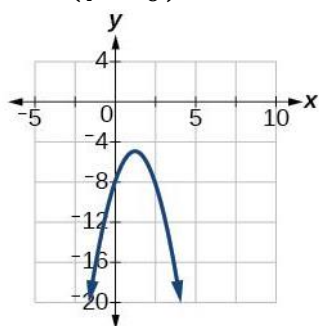
23. Vertex: $\left(\frac{5}{2}, -\frac{49}{4}\right)$; y-intercept: $(0, -6)$; x-intercepts: $(6, 0), (-1, 0)$



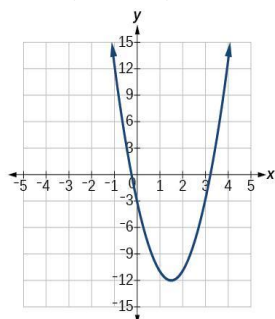
24. Vertex: $\left(\frac{7}{2}, -\frac{37}{4}\right)$; y-intercept: $(0, 3)$; x-intercepts: $\left(\frac{7 \pm \sqrt{37}}{2}, 0\right)$



25. Vertex: $\left(\frac{5}{4}, -\frac{39}{8}\right)$; y-intercept: $(0, -8)$; x-intercept: None



26. Vertex: $\left(\frac{3}{2}, -12\right)$; y-intercept: $(0, -3)$; x-intercepts: $\left(\frac{3 \pm 2\sqrt{3}}{2}, 0\right)$



27. $f(x) = (x - 2)^2 - 3$

28. $f(x) = (x + 1)^2 + 2$

29. $f(x) = -2(x - 2)^2 + 7$

30. $f(x) = -3(x + 1)^2 + 2$

31. $f(x) = \frac{1}{2}(x - 3)^2 - 1$

32. $f(x) = -\frac{1}{4}(x + 2)^2 + 3$